

AS Security Profiles for Secure and Resilient Internet Paths

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Abstract

The security and resilience of Internet paths depend on the security of the ASes that forward traffic along those paths. While prior studies have measured various security properties of ASes, such as routing security and DDoS protection, these properties are typically considered in isolation. As a result, network operators, researchers, and policymakers lack a common way to reason about AS security and to leverage it for improving the security and resilience of the Internet paths. In this paper, we introduce the concept of *AS security profiles*, which describe the security properties of ASes in an integrated way through multiple observable security attributes and by considering both routing security and infrastructure security properties of an AS. We demonstrate the feasibility of this concept by showing that many security attributes can already be obtained through Internet-wide measurements and by discussing how such profiles could be deployed in practice. We further discuss how AS security profiles can support applications such as security-aware peering, security-aware path selection, and security services offered by Internet Exchange Points (IXPs), thereby contributing to a more resilient and responsible Internet. Finally, we identify key research challenges, including profile trustworthiness, maintenance and publication, privacy, and interoperability with existing Internet infrastructure.

CCS Concepts

• **Networks** → **Network measurement**.

Keywords

AS security profile, Security

ACM Reference Format:

Shyam Krishna Khadka, Suzan Bayhan, Ralph Holz, and Cristian Hesselman. 2026. AS Security Profiles for Secure and Resilient Internet Paths. In *First International Workshop on a Resilient, Renewable, and Responsible Internet (RE-NET '26)*, December 07-10, 2026, Utrecht, Netherlands. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3843201.3856201>

1 Introduction

The Internet is an interconnected collection of around 79k [10] networks, as of September 2026. They are known as Autonomous Systems (ASes) and they exchange routing information using the Border Gateway Protocol (BGP) [23], thereby providing Internet paths. Consequently, the security and resilience of Internet paths depend on the individual, uncoordinated security practices of the ASes that carry traffic along them. Given the Internet's critical role in modern infrastructure, this dependency on inconsistent AS-level security has become a significant and pressing concern [13].

Researchers have measured individual security properties of ASes, including routing security [9, 11, 29], DDoS resilience [12, 17, 18], and router security [16]. However, these efforts remain isolated, with each focusing on a specific security aspect. As a result, there is currently no common way to reason about the overall security of an AS.

The lack of transparency about AS security makes it difficult for network operators to incorporate security into operational decisions, such as peering and path selection. Improving the visibility of AS security properties can therefore support more informed decisions, encourage the adoption of security best practices, and ultimately contribute to a more resilient and responsible Internet [13].

In this paper, we introduce the concept of *AS security profiles*, which characterize the security properties of ASes in an integrated way through multiple observable security attributes and by considering both routing security and infrastructure security properties of the ASes. Rather than prescribing a fixed set of attributes, we argue that AS security profiles should evolve as new security requirements and Internet measurement techniques emerge. We discuss how such profiles can be constructed using Internet-scale measurements, explore their potential applications, and identify



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RE-NET '26, Utrecht, Netherlands

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ACM ISBN 979-8-4007-2981-2/2026/12

<https://doi.org/10.1145/3843201.3856201>

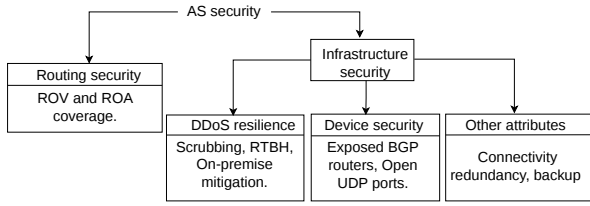


Figure 1: AS security represented in routing security and infrastructure security with example attributes.

the research challenges that must be addressed to make them practical.

The contributions of this paper are as follows:

- We introduce the concept of AS security profiles for describing the observable security characteristics of ASes, and demonstrate that many security attributes can already be obtained through existing Internet-wide measurements.
- We discuss how AS profiles can support applications such as security-aware peering, security-aware path selection, and security services offered by IXPs.
- We identify key research challenges, including profile trustworthiness, publication, privacy, and integration with existing Internet infrastructure, and outline directions for future research.

2 Security of Autonomous Systems

The security and resilience of Internet paths depend on the security of the constituent ASes on the paths. For the purpose of discussing AS security, we distinguish two complementary aspects: routing security and infrastructure security, as illustrated in Figure 1. These aspects address different security concerns and together contribute to the security characteristics of an AS as discussed next.

Routing security. The routing system governs how an AS exchanges and processes routing information with other ASes through BGP. Its security properties ensure that routing decisions are based on legitimate routing information and prevent the acceptance or propagation of unauthorized route advertisements. Otherwise, traffic may be redirected to unintended destinations or become unreachable due to route hijacks [1] or route leaks [26]. Well-known examples include the Pakistan YouTube incident, caused by a routing misconfiguration, and malicious BGP hijacks targeting cryptocurrency services [4]. Consequently, routing security is essential for preserving the *integrity* of routing information and the *availability* of Internet paths, both of which are fundamental to a resilient Internet.

Infrastructure security. Infrastructure security concerns the availability, robustness, and secure operation of the network infrastructure that supports an AS, including routers, switches, servers, and firewalls. Its security properties reduce the likelihood that attacks or device compromise disrupt the AS’s operation or affect traffic traversing it. Examples include DDoS resilience through scrubbing services or Remotely Triggered Black Hole (RTBH), device security through hardened routers, and other operational security practices. These properties contribute primarily to the *availability* and *resilience* of Internet paths.

Next, we describe how we can capture routing and infrastructure properties using observable security attributes.

3 ILLUSTRATIVE EXAMPLES OF Observable Security attributes

We make a first selection of observable security attributes by considering the example of security-aware path selection. That is a potential future form of routing in which ASes make routing decisions based on the routing and infrastructure security properties of ASes on potential paths, in addition to traditional routing metrics such as path length. This use case illustrates which security properties are already observable through Internet measurements today and how they can be combined into an AS security profile.

Table 1 summarizes the seven security attributes we identified for security-aware path selection: three related to routing and four to infrastructure security, including two concerning DDoS resilience and one concerning device security (see Figure 1). The table also shows the security role, example data sources, and the number of ASes we observed in each attribute as of September 2026.

The routing security attributes include Route Origin Validation (ROV), Route Origin Authorization (ROA) coverage, and forged-origin hijack involvement. These attributes contribute to the integrity of routing information by reducing the risk of route hijacks and route leaks.

The DDoS resilience attributes characterize an AS’s ability to mitigate attacks through scrubbing and RTBH [12]. In our example, the scrubbing measurement sources [16, 17] consider the top five scrubbers. The device security captures exposure through exposed BGP routers and open UDP ports.

Using the available measurement sources, we obtain the number of ASes observed by five of the seven attributes, excluding RTBH and forged-origin hijack involvement. For these two attributes, studies [12] and [14] provide specialized detection methodologies. However, we do not implement them because the required datasets are not readily accessible, and their detection involves large-scale data analysis that is not straightforward to incorporate into this work. For ROA coverage, Cloudflare Radar provides data at the per-AS

Table 1: Illustrative observable security attributes relevant to security-aware path selection.

Property	Attribute	Security role	Example source	Observed # of ASes
Routing security	Route Origin Validation (ROV)	Prevents acceptance of invalid route announcements (origin hijacks).	RoVista [24]	32,519
	RPKI ROA coverage	Enables origin validation and reduces the risk of prefix hijacking.	Cloudflare Radar [7]	Per-AS
	Forged origin hijack	Indicates ASes observed originating forged route announcements (negative indicator).	Methodology of [14]	No data available
DDoS resilience (Infra. security)	Scrubbing	Provides external mitigation through cloud-based scrubbing services.	Our methodologies [17, 18]	2,038
	Remotely Triggered Black Hole (RTBH)	Mitigates attacks by discarding traffic destined to attacked prefixes.	Methodology of [12]	No data available
Device security (Infra. security)	Exposed BGP routers	Indicates routers that accept unsolicited TCP connections (negative indicator).	Methodology of [16]	4,194
	Open UDP ports	Indicates exposure to UDP amplification attacks (negative indicator).	CyberGreen [8]	55,359

level rather than as a bulk-downloadable dataset. We observe substantial variation in coverage across sources: RoVista [24] covers 32k ASes, whereas the scrubbing based on the top five scrubbers [16, 17] identifies 2k ASes using scrubbing services. This variation highlights how data availability and granularity constrain the observable security attributes.

The attributes presented in Table 1 are sufficient to illustrate the security-aware path selection use case but are not intended to be exhaustive. Additional security attributes may become observable as new Internet measurement techniques emerge or when new applications that use security attributes emerge. For example, future work may investigate methods for identifying ASes that enforce maximum-prefix limits. Other security properties, such as internal monitoring capabilities or incident response procedures, are difficult to infer through external measurements alone and may require complementary evidence, such as operator disclosures, security audits, or regulatory reporting frameworks (e.g., NIS2 [21]).

4 AS security profiles

As we see from the previous sections 2 and 3, a single security mechanism cannot characterize the security of an AS. Instead, different users and applications often need to consider multiple security properties simultaneously. For example, a network operator evaluating a potential other AS to peer with may require information about both routing security practices of that AS and its operational resilience. We therefore propose organizing multiple observable security attributes into a common description of an AS, which we refer to as an *AS security profile*.

Table 2: Example AS security profile populated from existing measurement sources for AS32773.

Property	Attribute	Value
Routing security	ROV	100% [24]
	ROA coverage	0% [7]
	Forged origin hijack	No data available
DDoS resilience	Scrubbing	No [17, 18]
	RTBH	No data available
Device security	Exposed BGP routers	1 [16]
	Open UDP ports	4 [8]

Continuing the example of security-aware path selection, Table 2 shows an example of a security profile of AS32773. We populate the example profile using currently available measurements for five of the seven attributes.

As an AS security profile aggregates diverse security attributes, it utilizes heterogeneous data types. For instance, the profile in Table 2 represents the deployment of a traffic scrubbing service as a boolean value (i.e., “yes” or “no”) while it expresses the number of exposed BGP routers as an integer (e.g., a value of 1 indicates one routers within the AS have publicly accessible BGP ports).

AS security profiles enable ASes to select candidate paths in a rich way. For example, the profile of AS32773 (Table 2) enables ASes to compare the profile of AS32773 to those of other ASes in detail because AS32773 exposes multiple observable security attributes rather than reducing them to

a single score. For instance, an AS operator may prioritize routing-security attributes such as ROV to reduce the risk of routing attacks, while another deployment may assign greater importance to DDoS resilience or router security. Thus, AS security profiles provide a common set of observable security attributes based on which different applications can make their own assessments without requiring a universally agreed security score.

5 From concept to deployment

We now discuss how AS security profiles could be operationalized by building on existing Internet security initiatives. Table 1 shows that many security attributes can already be obtained through Internet-wide measurements or publicly available data sources. For example, we observed 55,359 ASes in CyberGreen and 2,038 ASes using scrubbing services. These observed values suggest that constructing AS security profiles is feasible using existing measurements. Such measurement platforms can continuously collect observable security attributes and organize them into a common representation of an AS's security posture.

One example of an operational model is the *Trust Zone* concept proposed by Clark et al. [6], in which groups of ASes agree to adopt routing security practices such as ROV, enabling customers to select providers that satisfy their security requirements. This demonstrates that security information can already be used to influence operational decisions in inter-domain routing. AS security profiles extend this idea by incorporating additional observable security properties beyond routing security, such as DDoS resilience and router security, thereby providing a more comprehensive view of an AS's security posture.

AS security profiles could be maintained by trusted third parties, such as IXPs, routing registries, or dedicated security organizations. These organizations already serve as coordination points for operational information and are well positioned to aggregate security attributes collected from Internet-wide measurement platforms and publicly available data sources, such as ROA and IRR records. The resulting profiles could be updated periodically to reflect changes in an AS's security posture and made accessible through standardized interfaces, such as application programming interfaces (APIs). This would allow network operators, researchers, regulators, and other applications to retrieve up-to-date security profiles consistently without requiring changes to existing routing protocols.

6 Applications

We discuss four major applications of AS security profiles that we envision:

1. *Security profile as a service.* IXPs, PeeringDB [22], or routing registries can offer a security profile as a service to their member ASes. For example, Servillo et. al [25] introduce the concept of a Risk Level-as-a-Service (RLaaS), a security service that IXPs can provide to their members, where an IXP assigns Risk Level (RL) dynamically to ASes by analyzing malicious announcements observed at its route server, potentially indicative of prefix hijacking [1]. IXPs could offer the same service with more security attributes and dimensions using AS security profiles. Similarly, IXPs could maintain and provide AS security profiles that incorporate multiple security attributes. Such a service would enable ASes to make security-aware peering decisions by selecting peers based on their security profiles.

2. *Security-aware transit and path selection.* AS security profiles can support their decision on selecting their transit provider and paths, alongside cost and performance. For routing, per-AS profiles can be aggregated into a path-level assessment. For example, [15] considers a minimum ROV score among ASes on the paths. This path-level score can then inform route preferences, while more advanced selection can evaluate multiple candidate paths using mechanisms such as BGP ADD-PATH [28].

3. *Continuous security assessment.* The security profiles of ASes can help them continuously assess how their networks appear from the perspective of external observers. Because the profiles are derived from external measurements, they can reveal inconsistencies between intended configurations and externally visible behavior, such as incomplete ROA deployment or exposed routers. In this way, AS security profiles complement internal security audits by providing an independent, Internet-facing view of the network. We expect this to become particularly important for compliance with cybersecurity regulations, such as NIS-2 [21].

4. *Policymaking and regulation.* Aggregated information derived from AS security profiles could support evidence-based policymaking at the regional or sector level. Existing Internet monitoring platforms (e.g., MANRS Observatory [20] and Basisbeveiliging [2]) could incorporate profile data to monitor the deployment of routing security and AS infrastructure resilience mechanisms. Rather than evaluating individual ASes, policymakers could use these aggregated indicators to identify systemic security gaps, track the adoption of security best practices, and assess the impact of cybersecurity policies over time. Such information could support initiatives such as the U.S. Roadmap for Enhancing Internet Routing Security [27] and the implementation of the NIS2 Directive.

7 Challenges & Research agenda

In Sections 3 and 5, we argue that many security attributes can already be measured and that AS security profiles can

build upon existing operational concepts such as Trust Zones. However, realizing AS security profiles at Internet scale requires addressing several technical and operational challenges. We view these challenges as a multi-disciplinary research agenda for the networking community rather than barriers to the concept itself and present five key challenges as follows:

1. Managing trustworthiness. The usefulness of AS security profiles depends on the correctness of the underlying measurements. Technical challenges include reducing false positives, improving measurement accuracy, verifiable logging of measurements, and validating profile attributes obtained from different data sources with potentially different measurement methodologies. However, trustworthiness is also an operational challenge because the organizations creating and publishing profiles and the ASes being evaluated may have different incentives. Future research should therefore investigate governance and business models that combine independent Internet-wide measurements with mechanisms for verification and operator feedback. Possible approaches include cross-validating results using multiple independent measurement platforms and allowing network operators to verify.

2. Maintaining and publishing. AS security profiles must be continuously updated as the security posture of ASes evolves. The Internet already relies on distributed operational information services, such as Internet Routing Registries (IRRs), PeeringDB, and RPKI repositories, to publish and synchronize network information, such as ROAs, across independent organizations. This suggests that AS security profiles could follow a similar operational model rather than requiring entirely new infrastructure. One possible direction is a distributed profile service operated by trusted organizations that periodically aggregates measurements from independent Internet measurement platforms and publishes updated AS security profiles through standardized interfaces. The research challenge is to ensure that independently operated repositories remain synchronized and provide timely, consistent profile information.

3. Incorporating in routing. The use of security profiles in routing can be an attractive use case, but may increase routing complexity. For example, security-aware path selection requires that BGP stores and forwards multiple paths to its peers and evaluates the path selection criteria, which could increase the size of the routing table. An important research question is how to incorporate security profiles into existing BGP-based routing systems while maintaining scalability and operational efficiency. One option could be to use a concept similar to a Trust Zone [6], where a group of ASes agrees to use security profiles in routing decisions. The ASes outside the group continue using traditional path selection criteria, such as cost and path length. As more ASes adopt

the profile framework to meet security requirements of their peers, the coverage of ASes using security profiles would expand across the Internet in an evolutionary way.

4. Managing incentives. The successful deployment of AS security profiles depends on creating incentives for operators to use and disclose them. If profiles are incorporated into operational applications such as security-aware routing and peering, and IXP services, they can influence routing and commercial decisions. This, in turn, gives operators a natural incentive to maintain accurate and up-to-date profiles. An important research question is how to design incentive mechanisms that encourage widespread participation.

5. Balancing privacy and transparency. While AS security profiles can improve AS transparency, publishing security information may also introduce risks, as adversaries could exploit it. However, not all profile attributes are equally sensitive. Routing security practices such as ROV deployment or ROA coverage are already publicly observable and are natural candidates for public disclosure, whereas detailed information about mitigation infrastructure or operational procedures may be shared only with authorized parties. This suggests that a layered disclosure model is feasible, where high-level security attributes are public while more detailed information is exchanged through trusted operational relationships, such as during peering negotiations or via IXP-managed platforms. The research challenge is to determine what security information should be publicly available and what should be restricted to trusted operational relationships, such as peering negotiations.

8 Related Work

Several initiatives have also proposed ways to characterize or improve aspects of AS security, but not in an integrated way across routing and AS infrastructure security, using our multi-attribute AS security profiles. MANRS [19] defines a set of operational practices for improving routing security, while Chang et al. [5] derive a routing-specific trust metric from BGP updates. Similarly, the CableLabs routing security profile [3] provides recommendations for deploying routing security mechanisms such as RPKI and ROV. These efforts demonstrate the value of describing ASes using security-related information, but remain largely focused on routing security.

Finally, systems such as CyberGreen [8] show that Internet-wide security reporting for ASes is operationally feasible by publishing security indicators based on open ports such as open NTP, SMTP, and chargen derived from Internet measurements. Together, these works suggest that many observable security attributes can already be measured and disseminated. This motivates the need to organize diverse

observable security attributes into a common AS security profile, rather than treating them in isolation.

9 Conclusions

This paper introduces the novel concept of an AS security profile, which characterizes the security properties of an AS through multiple observable security attributes. The profiles help increase the security and resilience of Internet paths because the AS operators have information about the security properties of ASes that their traffic will be forwarded to and help in using security properties in their routing decisions.

The security attributes discussed in this paper are intended as illustrative examples rather than an exhaustive set of attributes for describing the security properties of an AS. Additional attributes, such as security maintenance practices and incident management capabilities, may be incorporated as new security requirements and measurement techniques emerge.

Despite the usefulness of AS security profiles, various challenges need to be addressed for operationalization of this concept. Some of these challenges include managing the trustworthiness of the profiles and incorporating the profiles in routing. These challenges require cooperation among network operators, researchers, and internet registries.

Acknowledgments

This work was conducted as part of the projects CATRIN (www.catrinnl.nl) and UPIN (www.upin-project.nl), which received funding from the Dutch Research Council (NWO).

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